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AIM:Wireless Sensor Network Deployment and Coverage Analysis

Simulate random and grid deployment of sensor nodes and evaluate network coverage and connectivity.

1. **Grid deployment:** Nodes are placed in a 5 x 5 pattern at equal spacing.
2. **Random deployment:** Nodes are placed at uniformly random positions in the area.
3. **Coverage:** The area is divided into points 5 m apart. A point is covered if it lies within the sensing range of at least one node. Coverage % = covered points / total points x 100.
4. **Connectivity:** Two nodes are linked if the distance between them is within R_c . Breadth-first search (BFS) is used to find connected components.
5. Coverage and connectivity were calculated in the Tcl script from the node positions, and the topology was visualised in NAM.

```
set ns [new Simulator]

# Trace Files
set tracefile [open wsn.tr w]
$ns trace-all $tracefile

set namfile [open wsn.nam w]
$ns namtrace-all-wireless $namfile 500 500

# Topography
set topo [new Topography]
$topo load_flatgrid 500 500

# Simulation Parameters
set val(chan) Channel/WirelessChannel
set val(prop) Propagation/TwoRayGround
set val(netif) Phy/WirelessPhy
set val(mac) Mac/802_11
set val(ifq) Queue/DropTail/PriQueue
set val(ll) LL
set val(ant) Antenna/OmniAntenna
set val(ifqlen) 50
set val(nn) 9
set val(rp) AODV

create-god $val(nn)

# Configure Nodes
$ns node-config \
```

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```
-adhocRouting $val(rp) \  
-llType $val(ll) \  
-macType $val(mac) \  
-ifqType $val(ifq) \  
-ifqLen $val(ifqlen) \  
-antType $val(ant) \  
-propType $val(prop) \  
-phyType $val(netif) \  
-channelType $val(chan) \  
-topoInstance $topo \  
-agentTrace ON \  
-routerTrace ON \  
-macTrace ON  
  
# Create Nodes  
for {set i 0} {$i < $val(nn)} {incr i} {  
    set node($i)[ns node]  
}  
  
# =====  
# GRID DEPLOYMENT (3x3)  
# =====  
  
$node(0) set X_ 100  
$node(0) set Y_ 100  
  
$node(1) set X_ 200  
$node(1) set Y_ 100  
  
$node(2) set X_ 300  
$node(2) set Y_ 100  
  
$node(3) set X_ 100  
$node(3) set Y_ 200  
  
$node(4) set X_ 200  
$node(4) set Y_ 200  
  
$node(5) set X_ 300  
$node(5) set Y_ 200  
  
$node(6) set X_ 100
```

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```
$node(6) set Y_ 300

$node(7) set X_ 200
$node(7) set Y_ 300

$node(8) set X_ 300
$node(8) set Y_ 300

for {set i 0} {$i < $val(nn)} {incr i} {
    node(i) set Z_ 0
    $ns initial_node_pos node(i) 20
}

# =====
# Traffic
# =====

set udp0 [new Agent/UDP]
$ns attach-agent $node(0) $udp0
set null0 [new Agent/Null]
$ns attach-agent $node(8) $null0
$ns connect $udp0 $null0
set cbr0 [new Application/Traffic/CBR]
$cbr0 set packetSize_ 512
$cbr0 set interval_ 0.1
$cbr0 attach-agent $udp0
nsat1.0 "cbr0 start"
nsat8.0 "cbr0 stop"
# =====
# Finish Procedure
# =====

proc finish {} {
    global ns tracefile namfile
    $ns flush-trace
    close $tracefile
    close $namfile
    exec nam wsn.nam &
    exit 0
}
$ns at 10.0 "finish"
puts "Starting WSN Grid Deployment Simulation..."
$ns run
```

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```
sanjana@DESKTOP-N1QAMS3: ~/wsn$ code deploy.tcl
sanjana@DESKTOP-N1QAMS3: ~/wsn$ code deploy.tcl
sanjana@DESKTOP-N1QAMS3: ~/wsn$ ns deploy.tcl grid
num_nodes is set 25
warning: Please use -channel as shown in tcl/ex/wireless-mitf.tcl
INITIALIZE THE LIST xListHead
Mode: grid
Coverage      : 93.18 %
Total links   : 40
Connected components: 1
Largest component : 25 / 25 nodes
Network CONNECTED
```

OUTPUT:

